

Mohammadreza Nemati

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EXPERIENCE

AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Designing and deploying multi-agent systems with LLM-based reasoning for voice/text-driven HMI task automation using LangChain/LangGraph, developing evaluation frameworks with benchmarks for step completion, end-to-end execution, and grounding accuracy on real devices.
- Fine-tuning multimodal foundation models (Qwen3-VL-8B) using quantization-aware fine-tuning for agentic reasoning and grounding, performing model distillation from larger variants to optimize efficiency.
- Implemented agentic RAG system managing dynamic UI tree hierarchy for agent-based task planning.
- Deployed production LLM systems using vLLM for online model serving with continuous batching to improve throughput and latency.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Improved LLM KV-cache quantization by analyzing key/value norm imbalance; providing a theory-backed scheme that achieves 2× memory savings with more than 94% accuracy across models up to 70B [\[paper link\]](#)
- Developed LIT-LVM using latent variable models with structured regularization and GNN-based higher-order interactions for biomedical data, outperforming GNN, Factorization Machine, and elastic net baselines [\[paper link\]](#)
- Developed DEASurv, a deep survival model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve multi-modal biological survival prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Integrated vision transformers (SAM) into production mobile application with client-backend architecture, enabling real-time inference and achieving 7% IoU improvement through model optimization.
- Built multimodal system integrating vision-language models (LLaVA) with vision transformers for natural language-driven visual grounding and object segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 LLM for domain-specific applications using distributed training (FSDP) for multi-GPU setups and quantization for efficient deployment.

Machine Learning Research Assistant

08/2018 - 07/2022

University of Toledo

Toledo, Ohio

- Developed machine learning survival models for biological systems using over 500,000 patient records, applying advanced feature engineering and statistical modeling to predict clinical outcomes

SKILLS

- **Programming Languages:** Python, C++
- **Machine Learning:** Agent-based Systems, LLM-based Reasoning & Planning, Reinforcement Learning, LLM/VLM Fine-tuning, Vision Transformers, Graph Neural Networks
- **ML Infrastructure:** Model Deployment, Distributed Training (FSDP), Distributed Inference
- **Frameworks & Libraries:** PyTorch, LangChain/LangGraph, PyTorch Geometric (PyG), TensorFlow, sklearn
- **Inference Optimization:** vLLM, Quantization, Model Distillation
- **DevOps & Tools:** Docker, Flask, Redis, SQL, Git

EDUCATION

Case Western Reserve University

Cleveland, Ohio

Doctorate (PhD), Computer Science

08/2022 - 05/2025

University of Toledo

Toledo, Ohio

Masters (MS), Computer Science

08/2018 - 08/2020

Shiraz University

Shiraz, Fars

Bachelors (BS), Electrical Engineering

08/2012 - 05/2017